

Percussive Hemispheres: Turning the Head into a Resonating Instrument via Audio-Tactile and Visual Synchrony

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Figure 1: Percussive Hemispheres

Abstract

Percussive Hemispheres is an XR installation that explores the illusion that one's own head is functioning as a musical instrument. Participants sit in front of a half mirror, behind which a pair of color-emitting hemispheres appear visually fused with their head. When a steel drum and a corresponding point on the participant's head are struck simultaneously, the participant perceives the head itself as resonating—both visually and sonically. This illusion is often accompanied by a secondary effect, where the sound feels anchored to the head surface—a phenomenon we refer to as the Audiotactile magnet-head illusion. A laboratory study with 20 participants confirmed that synchronized head tapping significantly enhanced both the instrument-like sensation and the perception that sound was emerging from the head.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**.

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1 Introduction

Our research has focused on the surprising plasticity of how we perceive the interior of our own head—a perceptual domain that is rarely explored in mainstream XR. While the sensation of something existing inside the head has traditionally been regarded as abstract or metaphorical, our previous work has shown that such experiences can be concretely induced through multisensory integration. In *XRAYHEAD* and *XRAYHEAD Garden* [Imai et al. 2022][Kodaka and Kagami 2025], we demonstrated that combining synchronized visual, tactile, and auditory stimuli can generate a vivid illusion of an object penetrating into—or lingering inside—the participant's skull. Similarly, in *Being Mosquitone* [Kodaka et al. 2025], we employed intra-head auditory localization via headphones and HMD-based visual feedback to create the sensation of a mosquito flying around inside the head. These systems collectively suggest that it is possible to construct illusory head-internal space by tightly coordinating sensory cues, offering new possibilities for embodied XR design.

Building on this foundation, *Percussive Hemispheres* explores how synchronized auditory and tactile input can reshape the spatial perception of sound around the head. In our previous work (*XRAYHEAD Garden*), we demonstrated that coordinated audio-tactile stimulation could cause participants to perceive an external sound as originating from within their own head—a phenomenon we termed the Audiotactile magnet-head illusion [Kodaka and Kagami 2025]. In the present work, we extend this principle into a new

experiential domain: by striking a steel drum and the participant's head simultaneously at corresponding positions, participants begin to feel as if the hemispheres of their own head are resonating like an instrument. Rather than merely relocating the perceived origin of sound, the system transforms the head itself into a perceptual instrument, blending bodily self-perception with the acoustic properties of an external object. This shift—from hearing a sound in the head to becoming the sounding object—marks a novel direction in multisensory illusion design.

2 Implementation

This system inherits the core layout of the *XRAYHEAD setup* [Imai et al. 2022]. The participant faces a half mirror, which reflects their illuminated face while revealing a glowing object positioned behind the mirror. In the original *XRAYHEAD*, this glowing object was an illuminated skull model designed to appear as if it were embedded within the participant's semi-transparent head. In the present installation, we use a pair of color-emitting hemispheres as the glowing object in place of the anatomical skull. These hemispheres are handcrafted from washi paper and shaped to resemble the left and right cerebral hemispheres. Inside each hemisphere is a Philips Hue bulb, which is connected to a local wireless network. The bulbs are programmed to synchronize with the color values of a specific computer monitor, allowing each hemisphere to change color independently and dynamically.



Figure 2: Layout of the System

Behind the participant, separated by a black curtain, a steel drum is installed. The drum surface contains eleven distinct strike zones, each corresponding to a unique tone. Striking a specific zone triggers the associated hemisphere to emit a designated color. When the center zone is struck, both hemispheres simultaneously shift to different colors. The impact is detected by pressure sensors embedded beneath each strike zone. To induce the Audiotactile magnet-head effect, the experimenter simultaneously strikes the corresponding position on the back of the participant's head while hitting the steel drum. This synchronized stimulation creates a compelling perceptual illusion in which the external sound is experienced as if it were emanating from the participant's own head.

3 Evaluation

We conducted a laboratory study to evaluate the perceptual effects of *Percussive Hemispheres*. Twenty university or graduate students participated in the experiment. Each participant experienced two conditions in a within-subjects design. In both conditions, the visual stimulus—color-emitting hemispheres changing in synchrony with

the sound—was presented. In the Audio & Tactile condition, the steel drum and the participant's head were struck at corresponding positions simultaneously. In contrast, in the Audio-only condition, only the steel drum was struck, with no contact made to the head. After each condition, participants responded to two questions using a 7-point Likert scale ranging from -3 (not at all) to +3 (very strongly), with 0 indicating a neutral response. The first question asked, “Did it feel as if your head became an instrument?” The second asked, “Did it feel as if the sound came from your head?” — the latter serving as an indicator of the Audiotactile magnet-head effect.

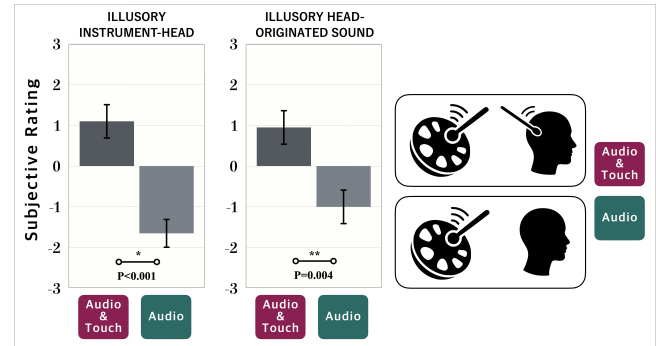


Figure 3: Experimental Results

As shown in Figure 3, both questions showed a significant shift from negative to positive ratings when comparing the Audio-only and Audio & Tactile conditions. Paired t-tests revealed statistically significant differences for both questions ($p < .001$ and $p = .004$, respectively). These results indicate that the addition of synchronized tactile input enhances the targeted perceptual illusions and support the effectiveness of the proposed system. In addition, many participants informally reported that their head felt materially altered, resembling the steel-drum surface itself (e.g., becoming harder, resonating with a metallic vibration, or evoking a sense of wearing an armor). This kind of material re-attribution parallels the *Marble-Hand Illusion* [Senna et al. 2014], in which tactile stimulation induces the sensation that one's own hand is made of marble. Such cross-modal transfer of material qualities suggests that the illusion we observed extends beyond mere localization, encompassing transformations of perceived bodily substance.

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